

Unit 6, Lesson 2: Using Models to Represent Ratios

Benchmark

MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b , or $a:b$ where $b \neq 0$.

Learning Targets

- ☐ I can write and interpret ratios.
- ☐ I can represent ratios using various models.

6.2.1 Warm-Up: Nana's Chocolate Milk

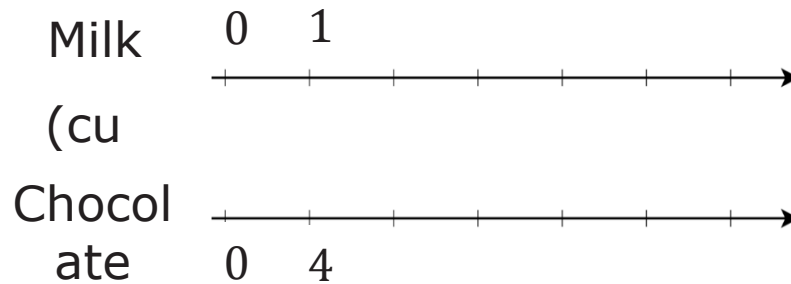
1. Dan's Nana prefers her chocolate milk made with just the right "chocolateyness," but he made a mistake when mixing it for her. The image below shows what Dan should have mixed and what he actually mixed.



- a. Describe what Dan did wrong.
- b. How could Dan fix it so that the glass contains Nana's preferred balance of milk and chocolate?

6.2.2 Exploration: Using Double Line Diagrams to Represent Ratios

Nana's chocolate milk recipe calls for 4 scoops of chocolate for every 1 cup of milk. One way to represent the recipe is with a double number line diagram.



1. Label each tick mark on the double number line diagram to represent the relationship between cups of milk and chocolate scoops.
2. How does the double number line show the ratio of Nana's preferred balance of milk and chocolate?
3. How many scoops of chocolate need to be mixed with 5 cups of milk to maintain the same ratio of milk to chocolate from Nana's recipe?
4. Circle where the answer to the previous question is represented on the double number line diagram.

Each pair of numbers on the double number line diagram represents an equivalent ratio. For example, the ratio 6 : 24 is equivalent to the ratio 2 : 8.

5. Complete the statement to interpret these two equivalent ratios in the context of Nana's recipe.

If 6 cups of milk are mixed with ____ scoops of chocolate, it will taste the same as mixing ____ cups of milk with 8 scoops of chocolate.

6. Recall Dan's error when mixing Nana's chocolate milk.

Nana's Recipe	What Dan Accidentally Mixed
1 cup milk 4 scoops chocolate	1 cup milk 5 scoops chocolate

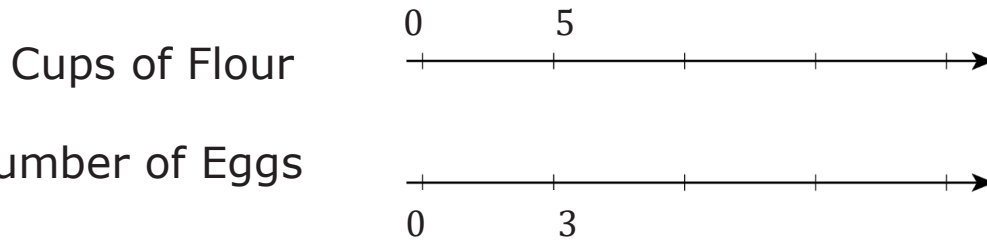
- a. Sabrina said Dan's mistake can be fixed by adding 0.5 cup of milk and 1 scoop of chocolate to the glass. Discuss with your partner whether or not you think Sabrina is correct.
- b. Complete the statement.

If Dan follows Sabrina's recommendation, the glass will contain ____ cups of milk and ____ scoops of chocolate.

7. Use the double number line diagram to determine whether Sabrina's recommendation will result in chocolate milk with Nana's preferred taste. If Sabrina's recommendation will work, mark the locations of this mixture on the double number line diagram with two Xs, one on each number line.

6.2.3 Try It: Using Double Line Diagrams to Represent and Interpret Ratios

1. A recipe for 1 batch of cookies calls for 5 cups of flour and 3 eggs. A double line diagram is shown to represent the recipe.



- a. What is the ratio of flour to eggs in 1 batch of cookies?
- b. Complete the diagram to show the amount of flour and eggs needed for 2, 3, and 4 batches of cookies.
- c. How many eggs and cups of flour are used in 4 batches of cookies?
- d. Complete the ratio comparing cups of flour to 6 eggs.
- e. Complete the ratio of eggs to 15 cups of flour.

$\square : 6$

$\frac{\square}{15}$

6.2.4 Guided Instruction: Using Tape Diagrams to Represent Ratios

A tape diagram is another way to represent ratios. All the parts of the diagram that are the same size have the same value, similar to increments on a number line.

The tape diagram shown represents the ratio of cats to dogs at a local animal shelter.

Number of Cats



Number of Dogs



1. Complete the statements using the tape diagram.

a. The ratio of dogs to cats is $\frac{\boxed{}}{\boxed{}}$.

b. For every ____ cats at the shelter, there are ____ dogs.

2. If there are 27 cats and dogs at the animal shelter in all, and the ratio of cats to dogs is 4:5, the same tape diagram can be used to represent the relationship.

Number of Cats



Number of Dogs

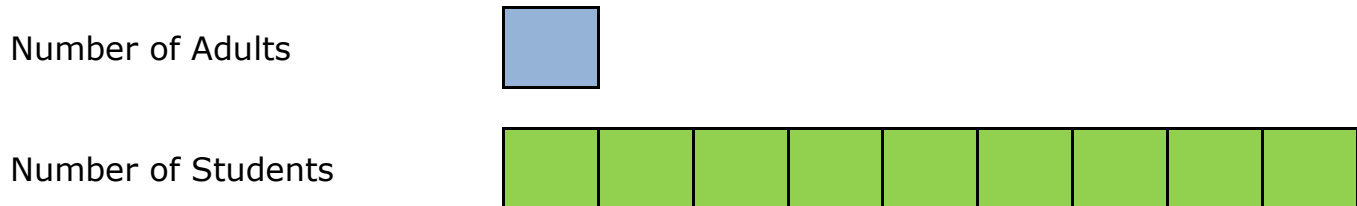


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- a. What is the value of each small rectangle in the diagram given this new information?
 - b. Label each small rectangle with this value.
 - c. How many dogs are at the animal shelter?
 - d. How many cats are at the animal shelter?
 - e. Complete the statement.

The ratio of cats to dogs at the shelter, 4:5, is equivalent to the ratio _____ to _____.

6.2.5 Your Turn!

1. Mr. Floyd is planning a field trip for his class to the Cade Museum for Creativity and Invention in Gainesville, Florida. To meet the museum's recommended ratio of chaperones to students, Mr. Floyd needs to have 1 adult chaperone for every 9 students. The tape diagram represents the ratio of adults to students needed.



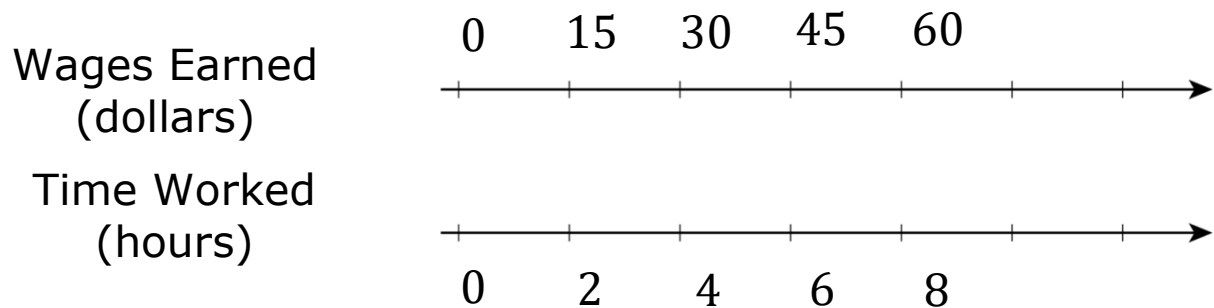
- a. What is the ratio of students to chaperones required by the museum?

There are 72 students who will attend the trip from Mr. Floyd's classes. Use the tape diagram to determine how many chaperones will be needed.

- b. Label each small rectangle with the number of people it represents.
- c. Complete the statement.

For all 72 students to attend, ____ chaperones are needed.

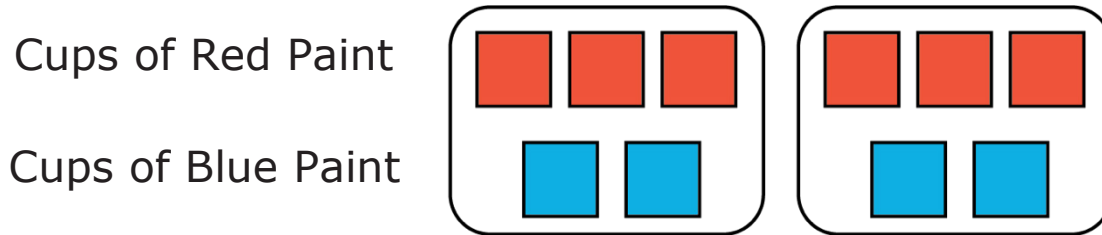
2. A cashier worked an 8-hour day and earned \$60.00. The double number line represents the amount she earned for working different numbers of hours.



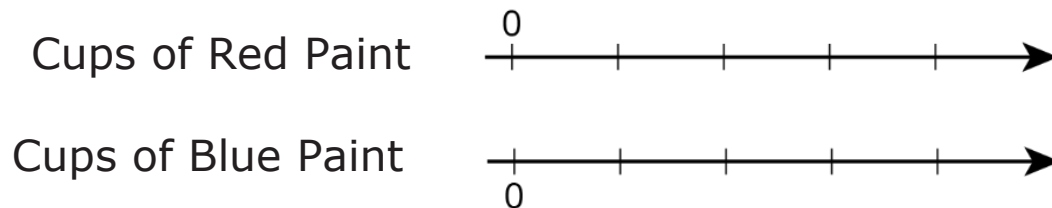
- a. Determine the amount of money she earned in 1 hour. Show your work and label it on the double number line.
- b. How much money does the cashier earn if she works 3 hours?

6.2.6 Wrap-Up: Ticket Out the Door

1. The representation shows the amount of red and blue paint that makes 2 batches of a particular shade of purple paint called lilac.



- a. On the double number line, label the tick marks to represent amounts of red and blue paint used to make batches of this shade of purple paint.



- b. How many batches are made with 12 cups of red paint?
- c. How much red paint should be mixed with 6 cups of blue paint to make lilac paint?
- d. Complete the statement.

To make this shade of lilac paint, mix ____ cups of blue paint for every ____ cups of red.